

BRENDAN TEO

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EDUCATION

University of California - Santa Cruz

October 2020 - June 2022

Bachelor's, Computer Science

- Software Engineering, Web Development, Analysis of Data Structures and Algorithms

PROJECTS & OUTSIDE EXPERIENCE

Pay-Per-Read

Santa Cruz, CA, USA

- Created a cutting-edge third-party web application that significantly enhances the experience of writers and bloggers posting on reputable news websites such as NY Times.
- Built and presented to the team with different UI/UX layouts that are concise and with the team's preference.
- Utilized React's virtual DOM and diffing algorithm in synergy to enhance the rendering speed of dynamic components on the webpage, effectively eliminating the need for page refreshes. This achieved a 40% decrease in page loading time.
- Conducted testing of database endpoints using Postman to verify the integrity of data exchanged between web pages.
- Modified website design led to a 25% increase in user time-on-page.
- Enhanced mobile responsiveness resulting in a 50% increase in mobile users' conversion rate.
- Implemented an Agile workflow, leading weekly meetings to facilitate discussions on progress and various coding approaches.
- Integrated the Stripe API with a Rust backend using the Rocket Framework that allowed credit card transactions with the end-user.
- [Link to project](#)

Slugstore

Santa Cruz, CA, USA

- Crafted a novel solution using the Py4Web framework to replace the existing online university bookstore by enhancing user experience and functionality.
- Engineered SQL server to efficiently handle HTTP requests to encapsulate several endpoints, leading to enhanced code readability and less time debugging.
- Implemented VueJS Router to enable dynamic client-side page routing, clearing auxiliary requests to the server and improving page loading speed.
- Programmed web pack for maintainability, and faster load times, and customized the configuration file to enable the hot module reloading.
- Executed manual functional and regression testing on features and builds to ensure their readiness for the final release.
- [Link to project](#)

Dihydrogen Monoxide Saving Gun

San Francisco, CA, USA

- Engineered a robotic system with the capability to enhance daily routines by dispensing water to plants and individuals.
- Integrated the OpenCV library into an Arduino-based robot, enabling it to identify facial attributes and objects.
- Utilized web-scraped training data to empower the robot's camera for accurate plant and facial pattern recognition, achieving an outstanding 90% precision in facial recognition assignments.
- [Link to project](#)

Multithread HTTP Server

- Developed a low-level implementation of an HTTP server in C, employing POSIX threads. It accommodates both single-threaded and multi-threaded models for managing incoming client connections and serving various HTTP requests such as GET, PUT, and HEAD.
- Ensured thread safety by integrating mutex locks with conditions on a queue data structure, establishing a secure environment for concurrent operations.
- Conducted comprehensive testing using cURL, evaluating server performance and reliability under both single-threaded and multi-threaded scenarios.
- [Link to project](#)

SKILLS

Programming Languages: HTML, CSS, JavaScript, Python, Java, C

Backend: NodeJS, ExpressJS, MongoDB, MySQL

Cloud: AWS, Google Cloud Platform

Testing: Selenium, Jenkins

Others: Git, Linux, Webpack, Docker, Agile(SCRUM), Jira

Languages: Mandarin, Cantonese, Japanese